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Simulation of partial melting material

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Simulation of partial melting material

by
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Dedication

To all future Longhorn graduate students.

Epigraph

What starts here changes the world.

—University of Texas at Austin

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Preface

Readers, this document represents a great deal of time and effort. Enjoy.
Cheers, Chenyu Tian.

Abstract

Simulation of partial melting material

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Chapter 1: Governing Equations in Mantle Dynamics

1.1 Mechanics of Mantle Dynamics

Fluid dynamics of the mixture of two phase flow has been extensively discussed. We will discuss the three governing equations of mass and momentum referring to the similar discussion in Drew and Segel (1971), McKenzie (1984) and Katz (2022). Considering a conservation equation for given volumetric variable γ in a representative volume element V .

$$\frac{d}{dt} \int_V \gamma d\mathbf{v} = - \int_{\partial V} \mathbf{f} \cdot d\boldsymbol{\sigma} + \int_V S d\mathbf{v} \quad (1.1)$$

where $\mathbf{f}$ represents flux travelling across the boundary of a representative volume element V . The minus sign indicates that the positive directions aligns with normal unit vector pointing outwards. S , source/sink term, represents the volumetric production rate of γ . We can apply Gauss's theorem to obtain differential form,

$$\frac{\partial \gamma}{\partial t} + \nabla \cdot \mathbf{f} = S \quad (1.2)$$

We also define the quantity of phase fraction rigorously with the help of an indicator function admitting that multiple phase cannot coexist at the same spatial point, which are regarded as immiscible phases. We define indicator function,

$$i(\mathbf{x}) = \begin{cases} 1 & \text{phase}_i \\ 0 & \text{not phase}_i \end{cases} \quad (1.3)$$

We then define macroscopic phase fraction function:

$$\phi_i = \frac{1}{\delta \mathbf{v}} \int_{\delta \mathbf{v}} i(\mathbf{x}) d\mathbf{v} \quad (1.4)$$

In the beginning of my discussion, We will discuss two immiscible phases liquid magma and solid mantle. Phase fraction of liquid magma is labelled as ϕ_l and phase fraction of solid mantle is labelled as ϕ_s , where $\phi_l + \phi_s = 1$ is always attained. Phase fraction

can also be regarded as volume fraction in macroscopic sense. Substituting bulk density $\bar{\rho} = \phi_l \rho_l + \phi_s \rho_s$ for generalized volumetric variable γ in equation 1.1, we can obtain conservation law for bulk mass,

$$\frac{\partial \bar{\rho}}{\partial t} + \nabla \cdot \bar{\rho} \mathbf{v} = 0 \quad (1.5)$$

with the help of Boussinesq approximation ignoring compressibility of each individual phase, and extend Boussinesq approximation further applying to density difference between each phase except body force term. We can write continuous function for two phase mixture simply as,

$$\nabla \cdot \bar{\mathbf{v}} = 0 \quad (1.6)$$

Substituting phase averaged momentum $\bar{\rho} \bar{\mathbf{v}}$, We then write conservation of momentum with phase averaged term,

$$\frac{\partial \bar{\rho} \bar{\mathbf{v}}}{\partial t} + \nabla \cdot \bar{\rho} \mathbf{v} \otimes \mathbf{v} = \bar{\rho} \mathbf{g} + \nabla \cdot \bar{\boldsymbol{\tau}} - \nabla \bar{P} \quad (1.7)$$

where $\boldsymbol{\tau}$ represents deviatoric stress, Ignoring change of inertia term due to near zero Reynolds number in magma and mantle,

$$\bar{\rho} \mathbf{g} + \nabla \cdot \bar{\boldsymbol{\tau}} - \nabla \bar{P} = 0 \quad (1.8)$$

Insert more details later!

Combining equations, I finally arrive with the full flow mechanical system,

$$\phi_f (\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l} \nabla (p_l - \rho_f \mathbf{g}) = 0, \quad (1.9)$$

$$\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) = 0, \quad (1.10)$$

$$\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) = (\rho_s - \rho_l) \mathbf{g}, \quad (1.11)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \phi_f (p_l - p_s) = 0, \quad (1.12)$$

where the deviatoric stress of the mixture is

$$\tau(\mathbf{v}_s) = 2\mu_s\phi_s(\mathcal{D}\mathbf{v}_s - \frac{1}{3}\nabla \cdot \mathbf{v}_s\mathbf{I}) \quad (1.13)$$

and $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla\mathbf{v}_s + \nabla\mathbf{v}_s^T)$ is the symmetric gradient. Moreover, the relative permeability is $k_{\phi_l} = k_0\phi_l^{2+2\Theta}$, where $\Theta \in [0, 1/2]$. μ_s and μ_l are dynamic viscosities for solid and liquid correspondingly.

1.2 Conservation of Chemical Species and Energy

Chemical compositions of mantle and magma is very complicate and very hard to trace at the same time. We simplify the model to a two-phase system with two species. We negelect detailed description of chemical reaction process and effect on the Gibbs free energies. Similar discussion can be found in extensive literatures like Spiegelman and Elliott (1993), Hewitt and Fowler (2008), Katz (2008) and Marc A. Hesse and Oza (2022). Both two species can be solid or liquid. Two species are immiscible as solid but miscible as liquid. I consider a homogenous density ρ_s for solid phase and ρ_l for liquid phase. With extened Boussinesq approximation later assuming no density difference except in body force term, we can make further assumption that $\rho_l = \rho_s = \rho$ Within the representative volume element V , $\gamma_{i,j}$ represents one quantity

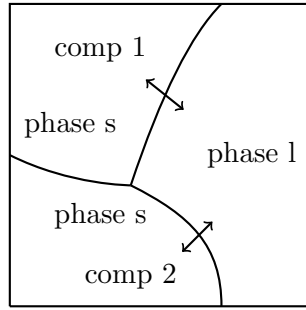


Figure 1.1: An representative volume element (RVE).

of the specie i presenting in phase j , where $i \in 1, 2$ and $j \in \{s, l\}$. The mass fraction $c_{i,j}$ of a given specie i in phase j is often regarded as concentration. Hence, we can

write transport equations for a individual specie i assuming a pseudo melting term Γ ,

$$\frac{\partial}{\partial t}(\phi_l \rho_l c_{i,l}) + \nabla \cdot (\phi_l \rho_l c_{i,l} \mathbf{v}_l - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}) = \Gamma_i \quad (1.14)$$

$$\frac{\partial}{\partial t}(\phi_s \rho_s c_{i,s}) + \nabla \cdot (\phi_s \rho_s c_{i,s} \mathbf{v}_s - \phi_s \rho_s \mathcal{D}_{i,s} \nabla c_{i,s}) = -\Gamma_i \quad (1.15)$$

where we can safely get rid of dissusion term in solid phase. Summing up equation 1.14 and equation 1.15, we can write a advection-diffusion equation in phase averaged volumetric bulk variable,

$$\frac{\partial \overline{\rho c_i}}{\partial t} + \nabla \cdot (\overline{\rho c_i} \mathbf{v} - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_i) = 0 \quad (1.16)$$

where the phase averaged volumetric bulk variable are defined as $\overline{\rho c_i} = \sum_{j \in \{s,l\}} \phi_j \rho_j c_{i,j}$ and $\overline{\rho c_i} \mathbf{v} = \sum_{j \in \{s,l\}} \phi_j \rho_j c_{i,j} \mathbf{v}_j$. The subscript i will be omitted since only one chemical specie is traced in this two-component system. Extending Boussinesq approximation $\rho_l = \rho_s = \bar{\rho}$, the specie transport equation can be simplified as,

$$\frac{\partial \bar{c}}{\partial t} + \nabla \cdot (\bar{c} \mathbf{v} - \phi_l \mathcal{D}_l \nabla c) = 0 \quad (1.17)$$

We write energy balance equation in the phase-averaged internal energy form first,

$$\frac{\partial \overline{\rho u}}{\partial t} + \nabla \cdot \overline{\rho u \mathbf{v}} - \nabla \cdot \overline{K \nabla T} = \overline{\rho \mathcal{Q}} + \nabla \cdot \overline{\boldsymbol{\sigma} \cdot \mathbf{v}} + \overline{\rho \mathbf{v} \cdot \mathbf{g}} \quad (1.18)$$

On the left hand side, we have time dependent rate of internal energy change, advection term and diffusion term. On the right hand side, we have heating/cooling source $\mathcal{Q}$, work exerted by stress and work against gravity g . In a thermochemical equilibrium scenario, we are replacing internal energy with enthalpy. We can write energy balancing equation neglecting additional heating/cooling term $\mathcal{Q}$ and dissipative heating McKenzie (1984),

$$\frac{\partial \overline{\rho h}}{\partial t} + \rho c_p \nabla \cdot \bar{\mathbf{v}} T = \rho L \nabla \cdot \phi_s \mathbf{v}_s + \overline{K \nabla^2 T} + \rho \alpha_\rho T \bar{\mathbf{v}} \cdot \mathbf{g} \quad (1.19)$$

Similar but slightly altered form was also discussed in Katz (2008).

1.2.1 Effective Velocity

Effective velocity in equilibrium transport for concentration within liquid phase is discussed in Spiegelman (1996). We also introduce our effective velocity, but instead of transporting concentration in liquid(magma), we transport bulk concentration. Define partition coefficient $D_i = c_{i,s}/c_{i,l}$ and

$$(\phi_l + \phi_s D_i) c_{i,l} = \bar{c}_i \quad (1.20)$$

$$(\phi_l / D_i + \phi_s) c_{i,s} = \bar{c}_i \quad (1.21)$$

Substitute $c_{i,l}$ and $c_{i,s}$ with equation 1.19 and equation 1.20 into bulk velocity $\bar{c}_i \bar{\mathbf{v}} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l}$,

$$\bar{c}_i \bar{\mathbf{v}} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l} = \left(\frac{\phi_l \mathbf{v}_l + D \phi_s \mathbf{v}_s}{\phi_l + \phi_s D} \right) \bar{c}_i \quad (1.22)$$

Hence, we get the proper expression for the effective velocity as $\mathbf{v}_{\text{eff}} = \frac{\phi_l \mathbf{v}_l + D \phi_s \mathbf{v}_s}{\phi_l + \phi_s D}$, where the partition coefficient D can be obtained through phase behavior package. If there is no melting $\phi_l = 0$, we can retrieve $\mathbf{v}_{\text{eff}} = \mathbf{v}_s$. If there is all melting $\phi_l = 1$, we can obtain $\mathbf{v}_{\text{eff}} = \mathbf{v}_l$. Hence, the definition of effective velocity is well defined in those two boundary scenarios.

1.3 Eutectic Phase Behavior

The enthalpy method requires an explicit relationship between temperature and enthalpy. Embracing the fact that Mid-ocean ridge forms at the place where temperature is generally lower, eutectic melting, where melting temperature is significantly lowered with the presence of the secondary specie, is a good model for modelling melting behavior at certain location. The binary eutectic phase behavior describes the chemical behavior of two immiscible crystals from a completely miscible solution. On a typical eutectic phase behavior diagram, we can locate any phase situation using enthalpy and composition information. I follow phase behavior details described in Marc A. Hesse and Oza (2022). Some adjustments and assumptions are made for our partially molten mantle system.

- The composition of the second specie in the liquid formed by eutectic melting X_e is a function of pressure.
- Melting temperature T_m is a function of pressure.

Following the previous discussion,

1.4 Rescaling and Nondimensionalization

Works Cited

- Donald Drew and Lee Segel. Averaged equations for two-phase flows. *Studies in Applied Mathematics*, page 3, 1971.
- I.J Hewitt and A.C Fowler. Partial melting in an upwelling mantle column. *Proc. R. Soc. A*, 464:2467–2491, 2008.
- Richard F. Katz. Magma dynamics with the enthalpy method: Benchmark solutions and magmatic focusing at mid-ocean ridges. *Journal of Petrology*, 49: 2099–2121, 2008.
- Richard F. Katz. *The Dynamics of Partially Molten Rock*. Princeton University Press, 1st edition, 2022. ISBN 9780691232645.
- Steve D. Vance Marc A. Hesse, Jacob S. Jordan and Apurva V. Oza. Supporting information for "downward oxidant transport through europa's ice shell by density-driven brine percolation". *Geophysical Research Letters*, 2022.
- Dan Mckenzie. The generation and compaction of partially molten rock. *Journal of Petrology*, 25:713–765, 1984.
- Marc Spiegelman. Geochemical consequences of melt transport in 2-d: The sensitivity of trace elements to mantle dynamics. *Earth and Planetary Science Letters*, pages 115–132, 1996.
- Marc Spiegelman and Tim Elliott. Consequences of melt transport for uranium series disequilibrium in young lavas. *Earth and Planetary Science Letters*, 118: 1–20, 1993.